

Nightmares Case 1: Bradycardia

Section 1: Case Summary

Scenario Title:	Nightmares Case 1: Bradycardia
Keywords:	Syncope, third-degree heart block, bradycardia
Brief Description of Case:	This is a case of an elderly patient with syncope. He is found to be in third degree heart block. The team is expected to perform an initial assessment and obtain an ECG. Upon recognizing the heart block, they should ensure iv access and place pacer pads while calling for help.

Goals and Objectives	
Educational Goal:	1. To develop an approach to the patient with syncope
Objectives: (Medical and CRM)	1. To prioritize initial investigations and therapies 2. To communicate a clinical scenario in a concise manner to the attending ER physician
EPAs Assessed:	To recognize an acutely unwell patient, call for appropriate help, and initiate a basic assessment and management plan.

Learners, Setting and Personnel			
Target Learners:	☐ Medical Students	☒ Junior Residents	☐ Senior Residents
	☐ Staff Physicians	☐ Inter-professional	☐ Other
Location:	☒ Sim Lab	☐ In Situ	☐ Other
Recommended Number of Facilitators:	Instructors: 1-2		
	Confederates: 1		
	Sim Techs: 1		

Scenario Development	
Date of Development:	2014
Scenario Developer(s):	Dr. Tim Chaplin
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Last Revision Date:	October 2019
Revised By:	Dr. Kyla Caners
Version Number:	2



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Robert White		Age: 82yo	Gender: Male		Weight: 90kg
Presenting complaint: Fall					
Temp: 37	HR: 27	BP: 80/50	RR: 18	O ₂ Sat: 99	FiO ₂ : room air
Triage note: Mr. White “fainted” while returning from the bathroom at home. He was found to be slightly more confused by his wife and complained of right elbow pain.					
Allergies: NKDA					
Past Medical History: Hypertension Coronary artery disease (MI in 2011) Anxiety Alcoholism Type-2 diabetes			Current Medications: Ramipril 10mg PO daily Metoprolol 100mg PO BID Metformin 1000mg PO BID Clonazepam 0.5mg PO TID ASA 81mg OD, Insulin humulin 30U qAM and 10U qhs, Septra 160mg PO BID.		

Section 2B: Extra Patient Information

A. Further History
<i>The patient (manikin’s voice): Your head feels a bit “cloudy” and you have elbow pain – you attribute this to the spine board. Otherwise unremarkable review of systems. You do not recall “passing out” at home.</i>
<i>The patient’s RN (confederate): If asked, can inform the learner that the patient’s wife described hearing a “thump” at home and then finding the patient on the floor. There were no signs/symptoms of a seizure.</i>
B. Physical Exam
<i>Other than bradycardia, there are no other significant findings No significant injuries (no signs of head trauma).</i>

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>). Adult, casual clothes, on a board and collar
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Cardiac monitors with pacing capability
C. Required Medications
Atropine, epinephrine infusion
D. Moulage
Elderly male in casual clothes on a backboard/C-spine collar.
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
Comfortable, a bit confused but pleasant. Can't recall "passing out at home" and head still feels a bit "cloudy".

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Bedside RN	You can assist with the history – unwitnessed collapse at home and ongoing mild confusion. There are no other pertinent findings on exam.



Simulation Scenario Template

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: 3 rd degree heart block HR: 28 BP: 80/50 RR: 18 O ₂ SAT: 99% T: 37°C GCS: 14	Mild confusion and amnestic to the event.	<u>Expected Learner Actions</u> ☐ Obtain a focused history ☐ Perform a focused examination ☐ Obtain a 12-lead ECG ☐ Obtain IV access ☐ Send labs (including electrolytes, troponin) ☐ Call staff MD for help	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - ECG completed → 2. Worsening Hypotension	
2. Worsening Hypotension HR: 25 BP: 75/45	Patient complaining of feeling more lightheaded, then is slower to answer questions.	<u>Expected Learner Actions</u> ☐ Initiate a fluid bolus ☐ Place pacing pads on patient and explain the process of transcutaneous pacing (consider analgesia) ☐ Trial atropine 0.5mg iv ☐ Trial epinephrine infusion ☐ Begin transcutaneous pacing	<u>Modifiers</u> - Atropine → no effect <u>Triggers</u> - End case PRN with handover to ER physician	

Simulation Scenario Template

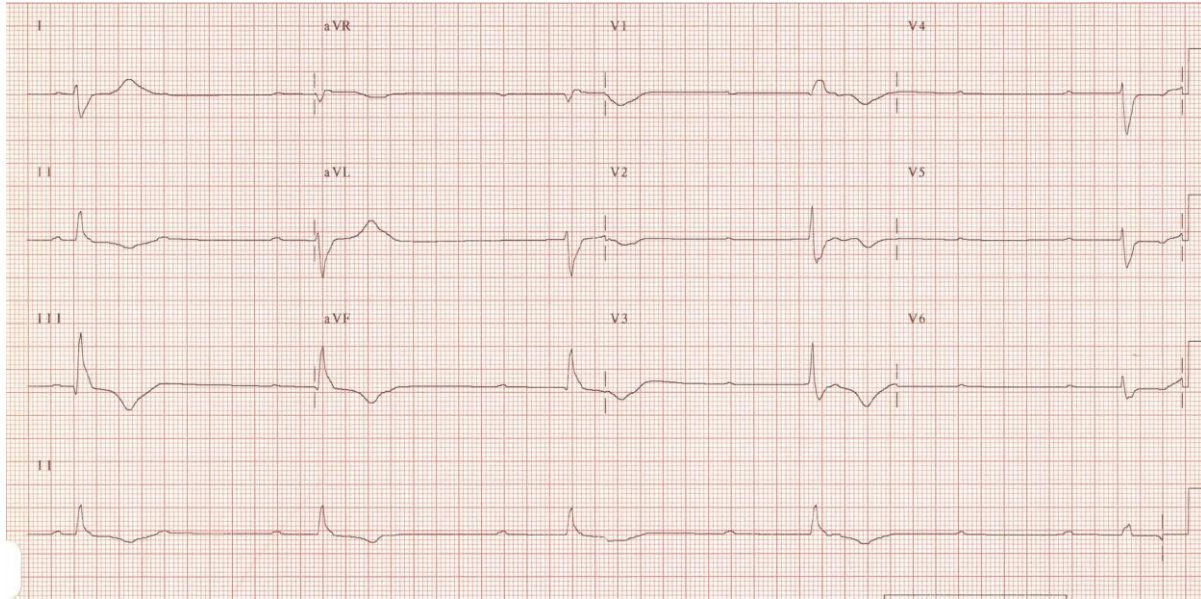
Appendix A: Laboratory Results – None Required for Case

<u>CBC</u> WBC Hgb Plt <u>Lytes</u> Na K Cl HCO ₃ AG Urea Cr Glucose <u>Extended Lytes</u> Ca Mg PO ₄ Albumin TSH <u>VBG</u> pH pCO ₂ pO ₂ HCO ₃ Lactate	<u>Cardiac/Coags</u> Trop D-dimer INR aPTT <u>Biliary</u> AST ALT GGT ALP Bili Lipase <u>Tox</u> EtOH ASA Tylenol Dig level Osmols <u>Other</u> B-HCG
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Simulation Scenario Template

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



ECG Source: <https://www.ecgquest.net/ecg/complete-heart-block-3/>

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issue expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

1. At the junior resident level, the expectation is to perform a focused history and physical examination, recognize profound bradycardia as the most likely cause for the patient's presentation, and initiate a basic plan. It is not expected that the junior resident independently perform transcutaneous pacing.
2. Review the AHA bradycardia algorithm:
<https://aclsalgorithms.com/wpcontent/uploads/2016/05/Website-Bradycardia-Algorithm-Diagram.pdf>

References

1. AHA Guidelines, 2015. <https://ahajournals.org/doi/10.1161/CIR.0000000000000261>

